

The Info Pharma Ordering System (IPOS) - Implementation Report



Centre for Software Reliability,
School of Informatics
City University,

London, 2026

Summary

This is the Implementation Report of the system created for the managing director of Info Pharma Ltd, by team 4, Readers of Systems Dependability at the Centre for Software Reliability, City University.

Authors

Herman Sildnes
Joanna Ayeni
Leon Seferi
Kanzah Hussain
Vladyslav Semanyshym
Yamin Ahmed
Jenis Khorshed

Distribution

The Managing Director of Info Pharma Ltd, the development team at the Centre for Software Reliability at the School of Informatics, City University London

Contents

1 Preface.....	3
1.1 Purpose and Scope of the document.....	3
1.2 Intended audience and Reading Suggestions.....	3
1.3 History of the document.....	3
2 Introduction.....	4
3 Glossary.....	5
4 User Requirements.....	6
5 System requirements.....	7
5.1 Purpose and scope of the system.....	7
5.2 System models.....	7
5.2.1 User stories.....	7
5.2.1.1 IPOS-SA-ACC.....	7
5.2.1.2 IPOS-SA-CAT.....	9
5.2.1.3 IPOS-SA-ORD.....	11
5.2.1.4 IPOS-SA-RPT.....	12
5.2.2 Prioritised List of User Stories.....	14
6 Anticipated Evolution of System Requirements.....	18
6.1 Validation.....	18
6.2 Enterprise SSO Login.....	18
7 Info Pharma High-Level Design.....	19
7.1 IPOS-SA.....	19
7.2 IPOS-CA.....	19
7.3 IPOS-PU.....	19
7.4 Software Architecture.....	19
7.4.1 Design diagram of subsystem IPOS-SA.....	20
7.5 Source Code (Java) for the interfaces of IPOS-SA.....	21
7.6 Testing the communications between subsystems.....	25
8 Appendices.....	31
8.1 Initial Statement of requirements by the client.....	31
8.2 Hardware (minimal and optimal configuration).....	31
8.2.1 minimal hardware configuration.....	31
8.2.2 optimal hardware configuration.....	31
9 Index.....	33

1 Preface

1.1 Purpose and Scope of the document

This is the Implementation Report for one of the subsystems of IPOS, IPOS-SA, and the Implementation of the IPOS software, which includes IPOS-SA and two other subsystems, IPOS-CA and IPOS-PU. The document covers the Software compilation and run-time components of the system as well as testing plans and results from subsequent tests on the system. The document is provided in response to the initial statement of requirements by Info Pharma Ltd, the client, for whom we then implemented this system for.

The document explains various features of the subsystem explaining what they should do and how the two other subsystems will interact with it. This document conveys in detail (using fragments of code and testing results) to the client the expected behaviour of the IPOS-SA subsystems and especially how this subsystem will interact with the other subsystems of IPOS.

1.2 Intended audience and Reading Suggestions

The Managing Director of Info Pharma Ltd, the system analysts and architects at the Centre of Software Reliability (CSR), City University London

Some of the sections are intended for the CSR team only, which is clearly stated in the document.

1.3 History of the document

This is v1.0 of the document.

2 Introduction

IPOS will allow Info Pharma Ltd to automate their business process in three interrelated areas: i) merchant account management, ii) ordering products and iii) selling inventory to the general public. By introducing an online ordering system the merchants will be able to make pharmaceutical orders and subsequently sell these products in their shops.

The three areas of planned automation are expected to be covered by three subsystems: IPOS-SA, IPOS-CA and IPOS-PU, respectively. This document is focused on IPOS-SA. The other subsystems are covered in separate documents.

In addition to the services offered to merchants, IPOS-SA allows Info Pharma to maintain an inventory of products which merchants can then purchase. Thus allowing merchants to directly buy products and configure discount plans for their purchases and set up inventory for their own business.

IPOS-SA will also send invoices and reports of products bought and profit generated when requested by the merchant.

IPOS-SA will interact with the other subsystems, IPOS-CA and IPOS-PU, too.

3 Glossary

CSR	Centre for software reliability
IPOS	Info Pharma Ordering System

4 Anticipated Evolution of System Requirements

We expect that IPOS-SA will be deployed quickly. The operational experience from merchants and staff will inform the future decisions about the necessary changes. The envisaged likely changes to various parts of IPOS-SA are summarised below.

4.1 Validation

Currently the validation of account information provided during registration of their account will be checked by administrators in Info Pharma. In the future, however, we envisage that the validation may be done using 3rd party online services, which will require changes.

4.2 Enterprise SSO Login

In the future, a feature we envisage for the system is having an enterprise SSO log in where employees of a merchant can use their SSO authentication to log into the system and access company specific data and products.

5 Software Compilation / run-time components

5.1 *Subsystem Set-up*

[basically the source files, programming languages used, compiler used]

[provide a project file description e.g. forms, data modules, etc]

5.1.1 *Files Created*

[a list of created binary files (.exe + .dll, etc files for implementation in python)]

[give each a short description and where they should be located at runtime (directories on target machine)]

[if more than one executable file is created a clear description should be given of who is to use them]

5.1.2 *source files*

[commented source files in zip file for online submission] - this section can be omitted we just need to upload the source files

6 Deployment of run-time software components on hardware

6.1 List of the COTS (Commercial-off-the-shelf) software components

[static libraries, DLLs, etc) which the application needs in order to run properly]

6.2 Deployment of run-time files on the target computer(s)

- if all components of our subsystem is deployed on the same computer deployment specification/diagram is not required - in this case an explicit statement must be included in the report that all run-time components are deployed on the same computer
- If a remote SQL server is used (on a different computer), a specification is necessary
- If we are using the executables of the other IPOS subsystems to check if our own prototype will access their data correctly, a deployment specification is required irrespective of if their subsystems are deployed all on one computer

7 Testing plans and testing results

7.1 Unit testing of IPOS-SA

7.1.1 testing plan

7.1.2 results

7.2 Subsystem testing with IPOS-PU and IPOS-CA

7.2.1 testing plan

7.2.2 results

8 Appendices

8.1 UML Design class diagram derived from final source code

[paste diagram here]

8.2 Initial Statement of requirements by the client

IPOS: Info Pharma Ordering System

The InfoPharma Ordering System (IPOS) will keep track of the orders of pharmaceutical goods made by merchants to InfoPharma Ltd and will assist the merchants' sales to consumers.

IPOS includes **three** subsystems:

- I. A server application (IPOS-SA), which in the fully functional production version will be administered by the IT department of InfoPharma Ltd.
- II. A client application (IPOS-CA), a copy of which will be run by the merchant on a desktop PC in the merchant's Pharmacy shop.
- III. An online platform (IPOS - PU) to be used by a merchant for direct sales of pharmaceutical goods to the public. The platform will allow the merchants to run advertisement/promotion campaigns with new products with discounts.

In the prototype of IPOS the three subsystems should be implemented as separate **desktop applications** able to interact with one another as necessary to meet the requirements. For instance, online sales (via IPOS - PU) should be linked to the inventory of the merchant: successful online purchases should be deducted from the stock of goods available at merchant's shop. Likewise, orders from IPOS-SA and delivered to a merchant, should be easy to record in IPOS-CA and increase the merchant's stock. Finally, promotion campaigns run via IPOS – PU may affect the cost of selling goods online (at reduced price), but not the cost of the same goods if sold in the pharmacy shops.

8.3 Hardware (minimal and optimal configuration)

8.3.1 minimal hardware configuration

The minimal hardware configuration recommended for IPOS development consists of a desktop PC, 1GB of RAM and at least 50GB of disk space for storing the software programs and data

possibly running a database server such as MySQL or other similar. A hosting package with preinstalled software and at least 1 GB for use data is also acceptable for both development and live deployment of IPOS.

8.3.2 optimal hardware configuration

The optimal hardware configuration envisaged for development should consist of a desktop PC, 4GB of RAM and at least 250GB of disk space for storing the software programs and data possibly running a database server such as PostgreSQL or better. Again, a hosting package with preinstalled software + 5GB of data is considered an optimal hardware configuration for both development and live deployment

9 Index

Not provided in this release of the document